GIT/GITHUB Questions for Cisco Automation 200-901

Theoretical (10)

1. **What's the purpose of git init**?
It is used for creating a new Git repository in a project folder.
It initializes version control by creating a hidden .git folder for tracking changes.

2. **How do you stage changes in Git**?
- git add <*filename1*> → stage a specific file
- git add . → stage all modified files

Preparing files to be included in the next commit

3. **What's the difference between git pull and git fetch**?

| | |
|---|---|
| git fetch | Downloads updates from remote repository but does NOT merge them |
| git pull | Downloads updates AND automatically merges them into your current branch |

4. **What does git status show**?
Current branch information / the Modified files (if any) / Staged files / Untracked files
It help one to understand the current state of the repository

5. **How do you create a new branch**?
git branch <*branch_name*>

6. **What's a Git merge conflict**?
This occurs when two branches of repository modify the same part of a file; git cannot automatically decide which change to keep

7. **How do you revert a commit**?
git revert <commit-id> - This method creates a new commit that reverses previous changes.
git reset --hard <commit-id> - Reset method (rewrites history)

8. **What's the purpose of .gitignore**?
.gitignore specifies files or folders Git should not track, such as: including temporary files (.tmp), password/config files (passwd, .conf), logs (.log), and build outputs

9. **How do you view commit history**?
git log --oneline , git log –oneline –graph –all , git log -p

10. **What's GitHub**?
It is a cloud-based platform that hosts Git repositories

Practical (10)

1. **Initialize a Git repo**: git init

```
root@labvm:/home/devasc# ls /
bin   devasc-vm-update-key.pub  EULA-Accepted  lib    libx32      mnt   root  snap      sys  var
boot  etc                       git_class      lib32  lost+found  opt   run   srv       tmp
dev   EULA                      home           lib64  media       proc  sbin  swapfile  usr
root@labvm:/home/devasc# man mv
root@labvm:/home/devasc# man mvdir
No manual entry for mvdir
root@labvm:/home/devasc# mv /git_class/ /home/devasc/
root@labvm:/home/devasc# ls
cisco  Desktop  Documents  Downloads  filelist.txt  git_class  git_tutorial  labs  snap  users.txt
root@labvm:/home/devasc# cd git_class/
root@labvm:/home/devasc/git_class# git status
fatal: not a git repository (or any of the parent directories): .git
root@labvm:/home/devasc/git_class# git init
Initialized empty Git repository in /home/devasc/git_class/.git/
root@labvm:/home/devasc/git_class# ls -al
total 12
drwxr-xr-x  3 root   root   4096 Feb 10 06:29 .
drwxr-xr-x 18 devasc devasc 4096 Feb 10 06:28 ..
drwxr-xr-x  7 root   root   4096 Feb 10 06:29 .git
```

2. **Clone a GitHub repo**: git clone <repo-url>

```
devasc@labvm:~/git_class$ git clone https://github.com/devhugo2057/git_class
fatal: could not create work tree dir 'git_class': Permission denied
devasc@labvm:~/git_class$ sudo su
root@labvm:/home/devasc/git_class# git clone https://github.com/devhugo2057/git_class
Cloning into 'git_class'...
Username for 'https://github.com': devhugo2057
Password for 'https://devhugo2057@github.com':
root@labvm:/home/devasc/git_class# git clone https://github.com/devhugo2057/git_class
Cloning into 'git_class'...
Username for 'https://github.com': devhugo2057
Password for 'https://devhugo2057@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/devhugo2057/git_class/'
root@labvm:/home/devasc/git_class# git clone git@github.com:devhugo2057/git_class.git
Cloning into 'git_class'...
remote: Enumerating objects: 28, done.
remote: Counting objects: 100% (28/28), done.
remote: Compressing objects: 100% (13/13), done.
remote: Total 28 (delta 7), reused 28 (delta 7), pack-reused 0 (from 0)
Receiving objects: 100% (28/28), done.
Resolving deltas: 100% (7/7), done.
root@labvm:/home/devasc/git_class# git branch
root@labvm:/home/devasc/git_class# ls
git_class
root@labvm:/home/devasc/git_class# cd git_class/
root@labvm:/home/devasc/git_class/git_class# git branch'
> ^C
root@labvm:/home/devasc/git_class/git_class# git branch
* master
root@labvm:/home/devasc/git_class/git_class# ls
DEVASC.txt  file2.txt  file.txt  python  README.txt
root@labvm:/home/devasc/git_class/git_class#
```

3. **Stage all changes**: git add .

```
root@labvm:/home/devasc/git_class/git_class# echo "Making mods in preparation for a staging action" >> DEVASC.txt && echo "Multip
le edits" >> file.txt && echo "All at once" >> file2.txt && echo "Update the README text file" >> README.txt
root@labvm:/home/devasc/git_class/git_class#
root@labvm:/home/devasc/git_class/git_class# git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   DEVASC.txt
        modified:   README.txt
        modified:   file.txt
        modified:   file2.txt

no changes added to commit (use "git add" and/or "git commit -a")
root@labvm:/home/devasc/git_class/git_class#
root@labvm:/home/devasc/git_class/git_class# git add .
root@labvm:/home/devasc/git_class/git_class#
root@labvm:/home/devasc/git_class/git_class# git status
On branch master
Your branch is up to date with 'origin/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   DEVASC.txt
        modified:   README.txt
        modified:   file.txt
        modified:   file2.txt

root@labvm:/home/devasc/git_class/git_class#
```

4. **Commit changes**: git commit -m "message"

```
root@labvm:/home/devasc/git_class/git_class# git commit -m "A couple of changes"
[master f9d5873] A couple of changes
 4 files changed, 4 insertions(+)
```

5. **Create and switch to new branch**: git checkout -b feature

```
root@labvm:/home/devasc/git_class/git_class# git checkout -b feature
Switched to a new branch 'feature'
root@labvm:/home/devasc/git_class/git_class# git branch
* feature
  master
root@labvm:/home/devasc/git_class/git_class#
```

6. **Push changes to GitHub**: git push origin main

```
root@labvm:/home/devasc/git_class/git_class# git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'git@github.com:devhugo2057/git_class.git'
root@labvm:/home/devasc/git_class/git_class#
root@labvm:/home/devasc/git_class/git_class# git push origin master
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 645 bytes | 645.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:devhugo2057/git_class.git
   093ee04..f9d5873  master -> master
root@labvm:/home/devasc/git_class/git_class#
```

7. **Pull changes from GitHub**: git pull

```
root@labvm:/home/devasc/git_class/git_class# git pull --all
Fetching origin
There is no tracking information for the current branch.
Please specify which branch you want to merge with.
See git-pull(1) for details.

    git pull <remote> <branch>

If you wish to set tracking information for this branch you can do so with:

    git branch --set-upstream-to=origin/<branch> feature

root@labvm:/home/devasc/git_class/git_class# git branch --set-upstream-to=origin/TestFeature
Branch 'feature' set up to track remote branch 'TestFeature' from 'origin'.
root@labvm:/home/devasc/git_class/git_class#
root@labvm:/home/devasc/git_class/git_class# git pull --all
Fetching origin
Already up to date.
root@labvm:/home/devasc/git_class/git_class#
```

8. **View changes**: git diff

```
root@labvm:/home/devasc/git_class/git_class# git diff
root@labvm:/home/devasc/git_class/git_class#
root@labvm:/home/devasc/git_class/git_class# git log --oneline
f9d5873 (HEAD -> master, origin/master, origin/HEAD, feature) A couple of changes
093ee04 Added .gitignore - sensitive data
b573b2b resolved merge conflict on new feature addittion
f52b3bc Adding a conflict from master
bfb1c34 (origin/TestFeature) Adding a conflict from the branch
0202939 Changes on README and DEVASC
f4f682d Included the README text file
0d906b2 initial commit
root@labvm:/home/devasc/git_class/git_class#
root@labvm:/home/devasc/git_class/git_class# git diff f9d5873 093ee04
diff --git a/DEVASC.txt b/DEVASC.txt
index 7469b26..df256fe 100644
--- a/DEVASC.txt
+++ b/DEVASC.txt
@@ -1,4 +1,3 @@
 Moving into GIT in Cisco DEVASC Automation class
 Automation is slowing taking off
 Testing branching
-Making mods in preparation for a staging action
diff --git a/README.txt b/README.txt
index 0e58602..8b41e40 100644
--- a/README.txt
+++ b/README.txt
@@ -2,4 +2,3 @@ Read Me File
 Updated README info
 This is the Test Feature conflict line

-Update the README text file
diff --git a/file.txt b/file.txt
index 76fdcec..bac8060 100644
--- a/file.txt
+++ b/file.txt
@@ -1,2 +1 @@
 Hello Worldgit status
-Multiple edits
diff --git a/file2.txt b/file2.txt
index fde3546..4ab7e6d 100644
--- a/file2.txt
+++ b/file2.txt
@@ -1,2 +1 @@
 This is file 2
-All at once
root@labvm:/home/devasc/git_class/git_class#
```

9. **Merge feature into main**: git merge feature

```
root@labvm:/home/devasc/git_class/git_class# echo "Including modification" >> DEVASC.txt && echo "Important Update" >> README.txt && git add README.txt DEVASC.txt
root@labvm:/home/devasc/git_class/git_class#
root@labvm:/home/devasc/git_class/git_class# git status
On branch feature
Your branch is ahead of 'origin/TestFeature' by 4 commits.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   DEVASC.txt
        modified:   README.txt

root@labvm:/home/devasc/git_class/git_class# git commit -m "Risky upgrades"
[feature 902285c] Risky upgrades
 2 files changed, 2 insertions(+)
root@labvm:/home/devasc/git_class/git_class# git status
On branch feature
Your branch is ahead of 'origin/TestFeature' by 5 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
root@labvm:/home/devasc/git_class/git_class# git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
root@labvm:/home/devasc/git_class/git_class#
root@labvm:/home/devasc/git_class/git_class# git merge feature
Updating f9d5873..902285c
Fast-forward
 DEVASC.txt | 1 +
 README.txt | 1 +
 2 files changed, 2 insertions(+)
root@labvm:/home/devasc/git_class/git_class#
```

10. **View commit log**: git log --oneline

```
root@labvm:/home/devasc/git_class/git_class# git log --oneline
902285c (HEAD -> master, feature) Risky upgrades
f9d5873 (origin/master, origin/HEAD) A couple of changes
093ee04 Added .gitignore - sensitive data
b573b2b resolved merge conflict on new feature addittion
f52b3bc Adding a conflict from master
bfb1c34 (origin/TestFeature) Adding a conflict from the branch
0202939 Changes on README and DEVASC
f4f682d Included the README text file
0d906b2 initial commit
root@labvm:/home/devasc/git_class/git_class#
```